CH 11

- forward fourier transform (continous)

$$X(jw) = \int_{-\infty}^{\infty} x(t) e^{-jwt}\, dt$$

- inverse fourier transform

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(jw)\, e^{jwt}\, dw$$

- ex) $x(t) = e^{-at} u(t)$ $\quad \therefore x(t) = \begin{cases} e^{-at} & t \geq 0 \\ 0 & t < 0 \end{cases}$

$$X(jw) = \int_{-\infty}^{\infty} x(t)\, e^{-jwt}\, dt$$

$$= \int_{0}^{\infty} e^{-at} e^{-jwt}\, dt = \int_{0}^{\infty} e^{-(a+jw)t}\, dt$$

$$= \left[-\frac{e^{-(a+jw)t}}{a+jw} \right]_0^{\infty}$$

$$= \lim_{t \to \infty} \left(-\frac{e^{-(a+jw)t}}{a+jw} \right) - \left(-\frac{e^{0}}{a+jw} \right)$$

$$= 0 - \left(-\frac{1}{a+jw} \right) = \frac{1}{a+jw}$$

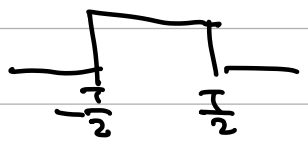
- $h(t) \rightarrow H(jw) = \frac{1}{a+jw}$ 이므로

$$|H(j\omega)| = \left|\frac{1}{a+j\omega}\right| = \frac{1}{\sqrt{a^2+\omega^2}}$$

$$\angle H(j\omega) = \frac{1\,(a-j\omega)}{(a+j\omega)(a-j\omega)} = \frac{a-j\omega}{a^2+\omega^2} = \frac{1}{a^2+\omega^2}(a-j\omega)$$

$$\arctan\left(\frac{-\omega}{a}\right) = -\arctan\left(\frac{\omega}{a}\right)$$

$$x(t) = \begin{cases} 1 & -\frac{T}{2} < t < \frac{T}{2} \\ 0 & \text{else} \end{cases}$$



$$= u\left(t+\frac{T}{2}\right) - u\left(t-\frac{T}{2}\right)$$



$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\,dt = \int_{-\frac{T}{2}}^{\frac{T}{2}} 1\cdot e^{-j\omega t}\,dt$$

$$= \left[\frac{e^{-j\omega t}}{-j\omega}\right]_{-\frac{T}{2}}^{\frac{T}{2}} = \frac{e^{-j\omega\frac{T}{2}} - e^{+j\omega\frac{T}{2}}}{-j\omega}$$

$$= \frac{e^{j\omega\frac{T}{2}} - e^{-j\omega\frac{T}{2}}}{j\omega} = \frac{2j\left(\frac{1}{2j}e^{j\omega\frac{T}{2}} - \frac{1}{2j}e^{-j\omega\frac{T}{2}}\right)}{j\omega}$$

$$= \frac{2j\,\sin\left(\omega\frac{T}{2}\right)}{j\omega} = \frac{\sin\left(\omega\frac{T}{2}\right)}{\frac{\omega}{2}}$$

$$\lim_{\omega\to 0}\frac{\sin\left(\omega\frac{T}{2}\right)}{\frac{\omega}{2}} = T$$

$\omega\frac{T}{2} = \pi$

when $\omega = \frac{2\pi}{T}$ 0 point 지남 ω.

- 예제 $X(jw) = \begin{cases} 1 & |w| < w_b \\ 0 & |w| > w_b \end{cases}$

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(jw)\, e^{jwt}\, dw = \frac{1}{2\pi}\int_{-w_b}^{w_b} 1 \cdot e^{jwt}\, dw$$

$$= \frac{1}{2\pi}\left[\frac{e^{jwt}}{jt}\right]_{-w_b}^{w_b} = \frac{1}{2\pi}\left(\frac{e^{jw_bt} - e^{-jw_bt}}{jt}\right)$$

$$= \frac{1}{2\pi}\left(\frac{2j\left(\frac{1}{2j}e^{jw_bt} - \frac{1}{2j}e^{-jw_bt}\right)}{jt}\right)$$

$$= \frac{1}{2\pi}\left(\frac{2j\,\sin(w_bt)}{jt}\right) = \frac{\sin(w_bt)}{\pi t}$$

$\lim_{t\to 0} \frac{\sin(t)}{t} = 1$ 이용.

$$= \frac{\sin(w_bt)}{w_bt} \cdot \frac{w_b}{\pi}, \quad \lim_{t\to 0} x(t) = \frac{w_b}{\pi}$$

$\sin(w_bt) = 0$

when

$w_bt = \pi \cdots t = \frac{\pi}{w_b}, \frac{2\pi}{w_b} \cdots$ 일 때 0